Gabarito - 2ª Av. - 2ª Chamada - Prob. Est. - Comp.

1 1.1 $P[x=5] = \binom{10}{5} 0{,}63^5 0{,}37^5 = \frac{10.9.8.7.6.5!}{5!.5.4.3.2} \cdot 0{,}63^5 \cdot 0{,}37^5$

$= 252 \cdot 0{,}63^5 \cdot 0{,}37^5 = 0{,}1734$

1.2 $P[X \leq 5] = 1 - P[X > 5] = 1 - [P[X=6] + P[X=7] + \cdots + P[X=10]]$

$= 1 - \left[\binom{10}{6} 0{,}63^6 0{,}37^4 + \binom{10}{7} 0{,}63^7 0{,}37^3 + \binom{10}{8} 0{,}63^8 0{,}37^2 + \binom{10}{9} 0{,}63^9 0{,}37 + 0{,}63^{10}\right]$

$= 1 - 0{,}63^6 \left[\frac{10.9.8.7.6!}{4.3.2.6!} \cdot 0{,}37^4 + \frac{10.9.8.7!}{3.2.7!} 0{,}63 \cdot 0{,}37^3 + \frac{10.9.8!}{2.8!} 0{,}63^2 \cdot 0{,}37^2 + \frac{10.9!}{9!} \cdot 0{,}63^3 \cdot 0{,}37 + 0{,}63^4\right]$

$= 1 - 0{,}63^6 [210 \cdot 0{,}37^4 + 120 \cdot 0{,}63 \cdot 0{,}37^3 + 45 \cdot 0{,}63^2 0{,}37^2 + 10 \cdot 0{,}63^3 \cdot 0{,}37 + 0{,}63^4]$

$= 1 - 0{,}63^6 [3{,}9357 + 3{,}8294 + 2{,}4451 + 0{,}9252 + 0{,}1575]$

$= 1 - 0{,}63^6 \cdot 11{,}2929 = 1 - 0{,}7061 = 0{,}2939$ //

$\mu = 63; \sigma = 4{,}828$

1.3 $P[X \leq 50] = P[-0{,}5 \leq X \leq 50{,}5] = P\left[\frac{-0{,}5-63}{4{,}828} \leq \frac{X-\mu}{\sigma} \leq \frac{50{,}5-63}{4{,}828}\right]$

$\approx P[-13{,}15 < Z < -2{,}59] = \Phi(-2{,}59) - \Phi(-13{,}15)$

$= 0{,}0038 - 0 = 0{,}0038$

2 2.1 X = "Número de meninos" $X \sim BN(3; 1/2)$

$E[X] = K\frac{(1-p)}{p} = 3 \cdot \frac{(1/2)}{(1/2)} = 3$

2.2 $P[X \geq 3] = 1 - P[X < 3] = 1 - [P[X=0] + P[X=1] + P[X=2]]$

$= 1 - \left[\binom{2}{2}(1/2)^0(1/2)^3 + \binom{3}{2}(1/2)^1(1/2)^3 + \binom{4}{2}(1/2)^2(1/2)^3\right]$

$= 1 - \left(\frac{1}{2}\right)^3\left(1 + 3 \cdot \frac{1}{2} + 6 \cdot \frac{1}{4}\right) = 1 - \frac{1}{8}(1 + 1{,}5 + 1{,}5) = 1 - \frac{4}{8} = \frac{1}{2}$

3

3.1 $E[X] = e^{\mu + \frac{\sigma^2}{2}} = e^{4 + \frac{0{,}64}{2}} = e^{4{,}32} = 75{,}1886$

$V(X) = \mu^2 e^{\sigma^2} - \mu^2 = (75{,}1886)^2 [e^{0{,}64} - 1] = (75{,}1886)^2 \cdot 0{,}8965$

$= 5068{,}2064$

$\Rightarrow DP[X] = 71{,}1913$

3.2 $P[X < 100] = P[\log X < \log 100] = P[Y < 4{,}6052]$

$= P\left[\frac{Y-\mu}{\sigma} < \frac{4{,}6052 - 4}{0{,}8}\right] = P\left[Z < \frac{0{,}6052}{0{,}8}\right] = \Phi[0{,}76]$

$= 0{,}7764.$

4 4.1 Área do círculo $= \pi$. fdp $= K$. Logo

$1 = \pi . K \Rightarrow K = 1/\pi \Rightarrow f(x,y) = \frac{1}{\pi}, \quad x^2 + y^2 \leq 1.$

Assim,

$f_X(x) = \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} 1/\pi \, dy = \frac{2}{\pi}\sqrt{1-x^2}, \quad -1 < x < 1$

$f_Y(y) = \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} \frac{1}{\pi} dx = \frac{2}{\pi}\sqrt{1-y^2}, \quad -1 < y < 1$

4.2 Temos,

$E[XY] = \int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} xy \, dy \, dx = \int_{-1}^{1} x \left[\frac{y^2}{2}\Big|_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\right] dx$

$= \int_{-1}^{1} \frac{x}{2}\left[1 - x^2 - (1-x^2)\right] dx = 0$

$E[X] = \int_{-1}^{1} \frac{2x}{\pi}\sqrt{1-x^2}\, dx = \int_{-1}^{0} \frac{2x}{\pi}\sqrt{1-x^2}\, dx + \int_{0}^{1} \frac{2x}{\pi}\sqrt{1-x^2}\, dx$

$= \int_{1}^{0} \frac{2u}{\pi}\sqrt{1-u^2}\, du + \int_{0}^{1} \frac{2x}{\pi}\sqrt{1-x^2}\, dx = 0.$

$\hookrightarrow u = -x \Rightarrow du = -dx$

Logo,

$Cov(X,Y) = 0 - 0 \cdot E(Y) = 0$

4.3 X e Y não são indep., pois $f_{X,Y}$ não fatora no produtos das marginais.